

ex:-

$$\star \{ X \in \mathcal{P}(\{1, 2, 3\}) : |X| \leq 1 \} = ?$$

$$\{ \{ \}, \{1\}, \{2\}, \{3\}, \{ \dots \} \}$$

$$|X| \leq 1 \quad \checkmark$$

$$\{ \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

2 2 2 3

$$\{ \{ \}, \{1\}, \{2\}, \{3\} \}$$

$$\{ \{ \}, \{1\}, \{2\}, \{3\} \}$$

$$\text{ex:- } \{ \{ X \in \mathcal{P}(A) : |X| \leq 1 \} \}$$

?

for any given set A where $|A| = m$.

→ if you have a set A with m elements, how many of them have cardinality of 0 or 1?

$$m+1$$

Homework.

Exercises of section 1.4.